

VGH Centre for Cardiovascular Health Cardiac Rehabilitation Program Introduction

6th floor, 2775 Laurel Street, Vancouver, V5Z 1M9
Telephone: 604-875-5389 Fax: 604-875-5794

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Welcome to the VGH Cardiac Rehabilitation program. Congratulations on your decision to participate! During your enrollment, we will be working with you to help you adopt healthy lifestyle habits to prevent or manage heart disease.

Through education and support, we will help you to **B.E.A.T.T.** heart disease. Our program goals are focused on:

- B**.ehaviours to help you take charge of your health and well-being
- E**.ating well and making healthy food choices
- A**.ctivities and exercise to improve your fitness and health
- T**.houghts and emotions that may be barriers to improving your life
- T**.ransition to independent activities or community supports/programs and resources where available.

But remember, **YOU** are the most important member of the cardiac rehabilitation team. Your commitment to the program is critical to your success.

During your enrollment, you will be working with a Case Manager regularly. Not only will your Case Manager be your main contact for all program related issues, they will also be your Health Coach to guide you through your health journey. Your program is also supported by a Registered Dietitian, Cardiac Counsellor/Social Worker, Psychiatrist, Program Cardiologists and a Clerical team.

As you may be participating in virtual activities, we would like to inform you that Virtual Care has some inherent privacy and security risks that your health information may be intercepted or unintentionally disclosed. In order to improve privacy and confidentiality, you should also take steps to participate in virtual care encounters in a private setting and should not use an employer's or someone else's computer/device as they may be able to access your information. For more information, please click on this link: <http://www.vch.ca/your-care/virtual-health>.

See Appendix A- Patient Letter and Notice of Using Video, Email and Texting to Communicate with your Health Care Provider

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Chapter 1: Your Physical Activity and Exercise Program

“Lack of activity destroys the good condition of every human being, while movement and methodical physical exercise save it and preserve it.”
Plato

The Ancient Greek philosopher, Plato, refers to movement and methodical physical exercise as necessary components to keep our bodies in good condition. During your cardiac rehab enrollment, you will work with a Case Manager who will help you develop strategies to be more active and encourage you to follow a structured exercise routine.

Physical activity, or movement, is anything you do throughout the day where your body is moving, such as light walking, standing, or general physical activities of daily living. In today’s society, we spend more time sitting throughout the day – we are spending less time moving our bodies. This type of *sedentary behaviour* is associated with increased risk of chronic disease. The association is so strong that you may have heard of the phrase, *“Sitting is the new Smoking”*.

Exercise, or methodical physical movement, is structured, more continuous, and is often performed at a certain intensity level. It involves training the body to maintain or improve its cardiovascular ability, muscular strength and endurance, flexibility and range of motion. Of course, improvements in performance can also be achieved through more specific-type of training, such as training for agility, speed, or power.

The important point to keep in mind is to always combine both, physical activity and exercise into your daily routines. Depending on your starting point, your Case Manager may work with you to first increase your physical activity – Move More, Sit Less strategies - before progressing you to a more structured routine through our weekly exercise program suggestions.

Your Weekly Exercise Program

Your weekly exercise program consists of 3 parts:



1. **Cardio (Aerobic) exercise:** This type of exercise increases your heart rate and breathing rate, and is usually done to a moderate-somewhat hard intensity. This type of exercise helps to strengthen your heart, lungs and muscles. The recommendation is to accumulate a minimum of 150 minutes per week, or at least 5 days per week for 30 minutes each day. Examples include walking, cycling, dance, aerobics, swimming, etc...
2. **Strength (Resistance) training:** This type of exercise gets you working with your own bodyweight, hand-held free weights, resistance bands or cans of soup! As we get older, or spend a lot of time in sedentary behaviours, our muscles lose strength and power. Regular strength training improves muscle strength, bone health, joint strength, balance, and reduces your risk of injuries and falls when performed properly. The recommendation is to do these types of exercises 2-3 days of the week.
3. **Flexibility (Stretching) and Mobility training:** Stretching is a great way to improve or maintain flexibility and range of motion (mobility). Best done at the end of your exercise to help cool your body down and to relax your muscles after a good workout. This type of exercise can be done every day. Other types of exercises you can perform daily include exercises to improve your balance and posture.

Exercise Program Options:

Depending on your needs and our program capacity, the following program options are available:

In-Person Exercise Program: You attend a weekly in-person class either at the hospital-based program (VGH Centre for Cardiovascular Health) or at one of our Happy Hearts Plus Community partner sites.

Virtual/Home-Based Program:

- You may require an initial period of in-person, introductory exercise and education classes before transitioning to your virtual/home-based exercise program.
- Virtual/Home-based weekly exercise programming in the form of on-demand exercise videos, graphic/printed exercise sets and once/week on-line, live exercise class using Zoom™;
- You must be a BC resident and physically be in the province of BC to access the virtual exercise classes.
- Access to internet and computer device recommended. Otherwise, telephone check-ins will be offered.

Hybrid Program:

Most participants will qualify to be enrolled in the Hybrid Program. This involves participating in a combination of in-person and virtual exercise and education before transitioning to solely virtual participation.

Most participants are ready to transition to their virtual program after a period of 1 to 3 months of in-person support.

Exercising Safely

Warm Up and Cool Down

Each time you start an exercise session, perform a 10 minute warm up to get your body ready for higher demands. This may simply be walking slowly, and adding some gentle movements such as shoulder and hip circles and dynamic stretching.

At the end of each exercise session, you should slow down to gradually allow your heart rate and breathing to come closer to your resting state. This may include walking slowly, practicing balance exercises, and stretching. If you don't take time to warm up or cool down, you are putting an increased stress on your heart and body, increasing your risk of feeling unwell or injuries.

Monitoring Intensity

These two simple ways can help you check if you are exercising too light, just right, or too hard.

Rating of Perceived Exertion (RPE) Scale: This is a measure of how hard you feel you are exercising on a scale of 0 to 10. You should aim for a 2 (easy) for warm up, 3-5 (moderate-somewhat hard) for your cardio or “conditioning” phase, and 2 (easy) for cool down. Listen to your body.

Talk Test: The easiest way to monitor your intensity. Talk to someone, or talk to yourself! If you can sing, it is very light intensity. If you can carry on a simple conversation without gasping for air, this is the right intensity. If you are gasping for breath, unable to speak → slow down, as you are working too hard!

Exercise heart rates can be determined by a simple calculation by using your maximum age predicted heart rate or from the results of your exercise stress test. Your Case Manager will help determine your exercise target heart rate, if applicable.

See Appendix B - ACSM (American College of Sports Medicine) Exercise Intensity Infographic

The F.I.T.T. Principle

This principle breaks down your exercise program as follows:

F.requency	How many days per week?	Public Health recommends that you exercise at least <i>5 days per week</i> or daily.
I.ntensity	How hard?	Aim for a <i>moderate-somewhat hard</i> effort. Your body will only physiologically adapt to exercise if it is challenged beyond your normal level activities.
T.ime	How long?	Public Health recommends that you exercise for <i>at least 30 minutes</i> each day. This can be continuous or broken up into shorter sessions. You should aim for at least 150 minutes per week.
T.ype	What type?	Try to incorporate all <i>3 main types of exercises</i> : cardio (aerobic), strength, and flexibility.

The Canadian Society for Exercise Society makes recommendations on Canadian Activity Guidelines.

FOR OLDER ADULTS - 65 YEARS & OLDER

Guidelines



To achieve health benefits, and improve functional abilities, adults aged 65 years and older should accumulate at least 150 minutes of moderate- to vigorous-intensity aerobic physical activity per week, in bouts of 10 minutes or more.



It is also beneficial to add muscle and bone strengthening activities using major muscle groups, at least 2 days per week.



Those with poor mobility should perform physical activities to enhance balance and prevent falls.



More physical activity provides greater health benefits.

General Tips for Strength Training



1. *Always warm up* your body with dynamic movements for 5-10 minutes; e.g. easy walk, light calisthenics and stretches.
2. *Never hold your breath!!* Do not bear down and hold your breath when lifting the weight. Breathe normally through-out the movement.
3. *Learn the movement and technique.* Practice the movement with good technique and posture without a weight or with a lighter weight. Once you feel you've mastered the technique, add or increase the weight to achieve moderate muscle fatigue with good technique.
4. *Control the movement:* count two seconds to lift the weight and 3 seconds to lower the weight.
5. Choose a *pusher* exercise (chest press, modified push-up), a *puller* exercise (seated row, lat. Pull-downs, band row), and a *leg* exercise (leg press, squat, split squat).
6. *If you can't do 8 (repetitions), decrease the weight.* You should be able to lift your chosen weight comfortably at least 8 times and achieve moderate muscle fatigue with 10-15 repetitions.

7. *Progress in small increments.* Start by adding 1-2 repetitions. Once you can do 15-17 reps, then try adding weight (the smallest increment available) and do 10-15 reps.
8. 1-2 sets of 10-15 repetitions is enough to see improvements in strength when done 2-3 times per week. Be sure to take a day in between strength workouts to recover.
9. *If there's pain, there is no gain!* If a movement is causing pain, try adjusting the weight, position and technique with the help of an exercise professional. If pain continues, stop doing the exercise and seek medical opinion.

See Appendix C - ACSM (American College of Sports Medicine) Resistance Training for Health Infographic

Normal “Expected” Responses to Exercise	Abnormal “Unusual” Responses to Exercise
Increased Breathing: It is normal to breathe faster and deeper when you are approaching moderate to somewhat hard exercise intensity.	Shortness of Breath: The feeling of not getting enough air, or gasping for breath, where you have a hard time catching your breath. Slow down, you may be working too hard!
Faster heart rate: to meet the demands of your body during exercise, your heart will pump faster and harder.	Irregular heart beats Chest pain, pressure, ache Discomfort down the arm, jaw, shoulders
Sweating: Your body’s way of regulating your body’s temperature.	“Cold Sweats”, Clammy
Sense of well-being, energized, and good mood	Nausea, Lightheadedness, Dizziness
You should continue exercising if you feel well.	STOP Exercise! If symptoms do not go away after 5 minutes of rest and you are feeling unwell, call 9-1-1. If you have nitrospray and you have chest pain, follow nitrospray protocol.

See Appendix D - Heart and Stroke Foundation Excerpt on Angina and Heart attack warning

See Appendix E - Exercise and Blood Sugar Monitoring for People with Diabetes

Finding a Place to Exercise at Home

We carefully curate exercise videos that are appropriate, home-based, and most suitable for cardiac rehabilitation. You may supplement your weekly exercise with your favourite activity, ie. walking, cycling, gardening. Most of the exercise routines require little space and sometimes no equipment at all. If you do not have equipment, such as bands or weights, you can try to substitute with soup cans, milk jugs, or just do your best to find something you can use to 'lift'. If you do not have bands, simply redo the workout from a previous week. Often times, your own body weight, works just as well.

We recommend you wear indoor running shoes, non-slip socks or be barefoot when you exercise at home to prevent slipping. We also recommend you clear your area of obstacles that can be tripping hazards, such as small rugs or clutter in the area. You should have a sturdy chair for support.

If exercising alone, have your telephone nearby to call someone to assist you. If you are exercising with us over Zoom, we may call you to check-in on your well-being or we may call emergency medical services to your home.

Chapter 2: CARDIAC HEALTH AND MEDICATIONS



Playing an active role in your heart health includes taking your medications as prescribed. If you have questions or concerns regarding your medications, please speak with your primary care providers or ask our medical team for some guidance.

You should always consult a health professional before stopping or changing your medications. Remember that your pharmacist is an excellent resource. You can also call 8-1-1, BC HealthLink to access pharmacy advice.

- Know the names of your medications and it is always a good idea to keep a list of your medications with you.

- Understand why you are taking the medication, be aware of main side effects and contact your doctor right away if you having side effects.
- Have a routine for taking your medications. Use a pill box or a schedule to remind you to take your medications
- Take your time. Double-check the name and dosage of all your medications before using them.
- Do not wait until you are completely out of medication before filling your prescriptions.

Chapter 3: HEALTHY HEART NUTRITION

A healthy diet, where diet means “a way of eating or pattern of eating”, such as how, when, what you eat, can help manage and reduce your risk of heart disease. What and how you eat can affect your cardiac risk factors, such as blood pressure, cholesterol, blood sugars and your weight. A healthy diet comprises of a balance of whole, natural foods with little to no foods that are highly processed. It is important to understand how your diet may be affecting your health. Our Program Dietitian is available to give you individualized nutrition counselling. Speak to your Case Manager to arrange an appointment.



Our Program Dietitian can help you with:

- Read food labels on packaged foods so you can find out how much fat, fibre, sodium, and sugar you are eating
- Incorporate more plant-based foods
- Add “Healthy Heart Optimizers” to your diet
- How to manage your weight
- How to manage your sodium and fluid intake if you have heart failure
- How to manage your blood pressure, cholesterol and blood sugars

- How to manage other conditions that may be affected by how and what you eat
- How to balance your plate
- Mindful eating habits

See Appendix F – CardioSmart, American College of Cardiology Heart Healthy Diet Infographic

Chapter 4: TAKING CARE OF MENTAL AND EMOTIONAL HEALTH



Our program offers emotional and mental wellness support through our Psychiatrist and Cardiac Counsellor. Heart disease affects us all differently, but it is not uncommon to feel anxious, stressed and even depressed with a new diagnosis. Persistent problems with stress can affect your heart. It is important to learn coping

strategies such as meditation to help manage your stress. There may be medical issues and medications that may contribute to your mental and emotional state. Speak to your Case Manager to arrange a consultation with our Psychosocial Team.

Please note:

- Appointments may be virtual or in-person
- Due to high demand, there is a waitlist. If you are not able to see one of our Psychosocial professionals during your enrollment, the program will provide recommendations and suggestions for community outreach programs and resources.
- Psychosocial support may be offered as 1:1 or as group workshops.

Chapter 5: YOUR PERSONAL CARDIAC REHAB PLAN

Being an active member of your cardiac rehab team means that you are directly involved in setting your health goals, following up with your health care providers, learning new skills, and taking action.

Part of your Cardiac Rehab program requires you to do some homework, such as reading materials, submitting activity logs, preparing for your appointments or simply attending your exercise sessions regularly. We encourage you to think about your health goals and set up a plan to achieve them.



Have you heard of **S.M.A.R.T.** Goals setting?

- Make your goals **S.pecific** Nail down only a few goals to work on at a time. Health and wellness goals may be broken up into 3 goals: 1 physical activity goal, 1 nutrition goal and 1 mental wellness goal.
- Make your goals **M.easurable** Try to answer how often, how much, and when?
- Make an **A.action Plan** Your plan must be action-based. What do you need to do to make things happen? What resources or supports you have or need?
- Make your goals **R.ealistic** Running a marathon next week is unrealistic if you don't run. Walking 8,000 steps per day is realistic if you are already walking 7,000 steps per day.
- Make your goals **T.ime sensitive** When will you do this by? Without a timeline (deadline), it is easy to say "next week, next month, next year".

*"You'll never change your life until you change something you do daily.
The secret of your success is found in your daily routine"*

– John C. Maxwell

Chapter 6: PROGRAM POLICIES

By enrolling into the program, you are in agreement and in acceptance of our program policies.

Participant Responsibilities:

- Participate in the program on a regular basis including: attending your exercise sessions, completing suggested readings, forms, activity logs, checking in with your Case Manager, and attending appointments with your Cardiac Rehab Health Team members.
- We cannot guarantee make up sessions if you do not attend regularly. We have the right to end your program enrollment if you miss sessions often.
- Seek medical advice for any health problems, symptoms or injuries that may affect safety of and ability to participate in exercise. Report these to your Case Manager. Report any medication changes.
- You must be physically in the province of B.C. to access the virtual exercise classes and appointments.

No-Show Policy:

For your cardiac rehab program to be effective, we need to have regular communication with you. Failure to attend your in-person or virtual sessions, failure to respond to your case manager check-ins, whether in-person, by email or telephone, constitutes a “no-show” on our records. After three consecutive no-shows (ie. 3 contact attempts), you will be automatically discharged from the program.

Appointment Cancellation:

Please provide 24-hour notice for cancellations or rescheduling of appointments whenever possible. We will not rebook appointment(s) if you repeatedly cancel or no-show for them. When you miss appointments, it affects others who are waiting their turn for consultations.

Program Fee Policy:

Payment of program fees is a condition of your program enrollment. We rely on program fees and annual fundraisers. However, finances should not be a barrier for participation. We do not require proof of financial burden if a subsidy is requested. Community program fees are set by and collected by the individual community centre.



Using a Video Appointment to Communicate with your Health Care Provider

Dear Patient,

Vancouver Coastal Health (VCH) is working hard to provide patients with opportunities for a convenient and reliable virtual care experience through a video visit with their care provider, rather than traveling into the hospital. You have been selected as an appropriate candidate for a video visit for your appointment. If you prefer to have an in-person visit, please let us know so that we can arrange an in-person appointment instead.

To be successful in having a video visit, you will need the following:

- ✓ Email
- ✓ Tablet, smart phone or computer
- ✓ Access to Wi-Fi internet
- ✓ A private space for your conversation

If you do not have the above items, you may prefer to come into the office or hospital to meet with your healthcare provider.

In accordance with British Columbia's *Freedom of Information and Protection of Privacy Act*, we will collect your personal information, specifically your first and last name and email address, to confirm your identity and enable you to access virtual health visits. We will send the registration invitations, appointment reminder notifications, and survey links only to the email address you provided. Your information, including the care you receive, is kept completely confidential.

Our intention is to improve your access to healthcare services through the use of technology, and improve your overall healthcare experience. However, we recognize that with the use of new technologies, things may not always go as planned. As a result, please advise us of any issues or how we can assist you further, because we are committed to ensuring your experience with the virtual health visit is seamless and positive.

There are some small risks involved with having a video visit. Please see the next page for some potential challenges.

Using a Video Appointment to Communicate with your Health Care Provider

Vancouver Coastal Health (VCH) staff and physicians are committed to making care easy for you while protecting your privacy. We are using a video appointment internet application called Zoom, which can come with some risks or limitations

What you need to know:

- By participating in the video visit, you are also agreeing to the collection of your personal information by Zoom. Zoom collects personal information, such as your first name, last name, email address as well as your device, network and internet connection (for example, IP address) in order to provide their services. Zoom processes and stores this information on its servers in the United States as well as its worldwide affiliates. We encourage you to review Zoom's privacy policy to learn more about how Zoom collects, uses and discloses personal information. Please visit www.zoom.us/privacy.
- Sometimes, a video appointment isn't appropriate for care. We will only offer this type of visit if we believe that it is a good option for you.
- You do not have to have a video appointment. If you prefer to meet in person, let us know.
- Sometimes technology does not work, and there could be unexpected problems. We will do everything in our power to minimize any issues with technology.
- If for any reason the Zoom service does not work, your team will arrange an alternate appointment.
- All the rules that apply to the practice of medicine in the province of BC apply to the use of technology and video appointments (e.g. documentation in the health record).
- Your care team will inform you if any other person(s) can hear or see any part of the conversation before the session begins.
- For your video appointment, you will need suitable Wi-Fi or network access. VCH cannot cover any data service costs associated with video appointments. Wi-Fi networks in public spaces (e.g. coffee shops) may not have secure connections so we recommend that you use your home internet network or a secure alternative.
- Although security measures such as encryption is used to safeguard communications, "perfect security" does not exist on the internet and VCH cannot guarantee the security of the information you share over the internet.
- Like other mobile applications, the Zoom application requires permissions to access content on your phone to function. For example, permission to access the camera and microphone enables users to have the video appointment.

Contact Information

If you have any questions or concerns about your appointment or would prefer to book an in person visit, please inform us when we call you to book your appointment.

If you have any questions regarding privacy or security, please contact the VCH Information Privacy Office at: privacy@vch.ca

Emailing or Texting your Health Care Provider

Vancouver Coastal Health (VCH) and Providence Health Care (PHC) staff and physicians are committed to communicating with you about your care while protecting your privacy and personal information. Email and text communication however, comes with risks and limitations that you should be aware of.

What you need to know:

- Detailed clinical discussions, counselling or crisis-related support will not be provided by VCH or PHC staff or physicians through email or text messages.
- Emails and texts are not an appropriate substitute for in-person or over-the-telephone communication, clinical examinations or for attending the Emergency Department when needed.
- Personal information that you send in email or text messages could be at risk if an email account and/or device is compromised. It is your responsibility to protect your accounts and devices from inappropriate access and/or loss.
- There is no guarantee that VCH or PHC will receive or be able to respond to emails or texts in a timely manner; therefore, email and text messaging should not be relied upon in urgent or emergency situations.

If you send an email or text to a VCH or PHC staff member or physician expecting a reply and do not hear back within a reasonable time period, it is your responsibility to follow up with that person regarding your care and/or appointments.

Contact Information

We welcome your comments and questions regarding emailing and texting between patients and VCH and PHC care providers. If you have questions, please contact us at the following:

[VCH Information Privacy Office](#)

Phone: (604) 875-5568

Email: privacy@vch.ca

[PHC Information Access and Privacy Office](#)

Phone: (604) 806-8336

Email: privacy@providencehealth.bc.ca

Tips for Monitoring Aerobic Exercise Intensity

Substantial health benefits are gained when adults achieve 150-300 minutes per week of moderate intensity exercise, or 75-150 minutes of vigorous intensity exercise per week. Several tools and methods are used to monitor exercise intensity. Utilizing these methods help adults achieve physical activity goals.



Talk test is a way to gauge exercise intensity based on ability to carry on a conversation.



Steady conversation is associated with moderate intensity aerobic exercise.



Vigorous intensity is an exercise intensity where only a few words are sustainable.

Heart rate

can be monitored by using a wrist watch and chest strap or a smart watch.



- A percentage of maximal heart rate indicates intensity (%HRmax)
- Moderate intensity exercise is estimated at 65-75 %HRmax
- Vigorous exercise is 76-96 %HRmax
- Target HR = $(220 - \text{age}) \times \%HR_{\text{max}}$

Perceived effort

is a subjective method to monitor how hard exercise feels.

- Commonly reported as a rating of perceived exertion (RPE)
- Scale is a range between 0-10
- Moderate intensity exercise is targeting an RPE of 3-4
- Vigorous exercise is an RPE of 5-7

Rating	0	1	2	3	4	5	6	7	8	9	10
Descriptor	Rest	Very easy	Easy	Moderate	Somewhat hard	Hard	-	Very hard	-	Very, very hard	Maximal effort

Motion sensors

are devices used to track steps and other activities.

- Moderate intensity exercise is a step rate of 100 steps per minute, or 1,000 per 10 minutes
- A common recommendation is to achieve 3,000 steps in 30 minutes
- Vigorous intensity is > 100 steps per minute.



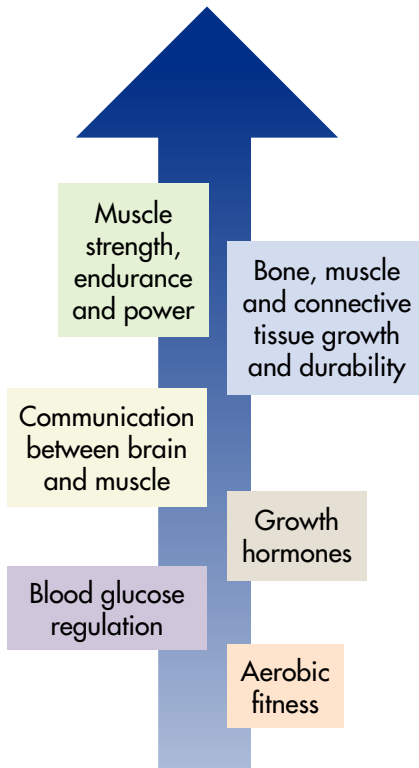
In addition to the above 0-10 scale, the Borg Scale of Perceived Exertion, which rates exertion on a scale of 6-20, can also be used.

Author: Micah Zuhl, Ph.D.; 2020

Resistance Training for Health

People of all ages and abilities who regularly participate in resistance exercise reduce risk of numerous diseases, improve quality of life and reduce mortality.

Key Physiological Benefits of Resistance Exercise



Resistance Exercise Can Help Manage and Treat Many Conditions Including:

- Arthritis
- Cancers
- Cardiovascular disease
- Dementia
- Depression
- Diabetes
- Fall risk
- Frailty
- Hypertension
- Insomnia
- Low back pain
- Mental health
- Movement disorders
- Obesity
- Osteoarthritis
- Osteoporosis
- Pulmonary disorders
- Peripheral vascular disease
- Stroke

Training can be time efficient and effective for health benefits:



For health benefits, muscles need to be challenged with a combination of weight lifted, repetitions and speed of lifting. The addition of resistance training to aerobic programs can also enhance other health gains throughout the life span from childhood to old age.

Exercise Plan:

- Free weights, machines and/or bands can be used
- Perform 8-10 multi-joint exercises that stress the major muscle groups
- Perform 2-3 sets of 8-12 repetitions with good form
- Lift and lower the weight in a controlled manner (2 seconds each up and down)
- The last repetition should be difficult to complete
- Perform exercise 2-3 times per week
- Progress weight lifted over time so that it feels like an 8 out of 10 difficulty (where 0 = no effort, 10 = hardest effort you can give)

What do I do if I have angina?

Rest or a medication called nitroglycerin (or nitro) usually helps angina. Nitroglycerin widens or opens up the coronary arteries. This allows more blood to reach your heart. You need to see a doctor if you are having angina. The doctor will prescribe nitroglycerin. Always carry it with you.

Stable angina	Unstable angina
• The same each time • Length of discomfort lasts for less than 15 minutes • Physical activity, exercise and/or emotional stress are common causes or triggers	• Less predictable • Length of discomfort lasts for more than 15 minutes • Can happen at any time, even during sleep • You feel chest pain you did not have before

What can I do to reduce my angina?

Avoid the four E's:

Exertion: too much physical activity

Eating a large meal

Emotional stress

Environment: too hot or too cold

Any of the four Es make your heart work harder. This means your heart needs more oxygen. If your heart does not get enough blood, angina can occur.

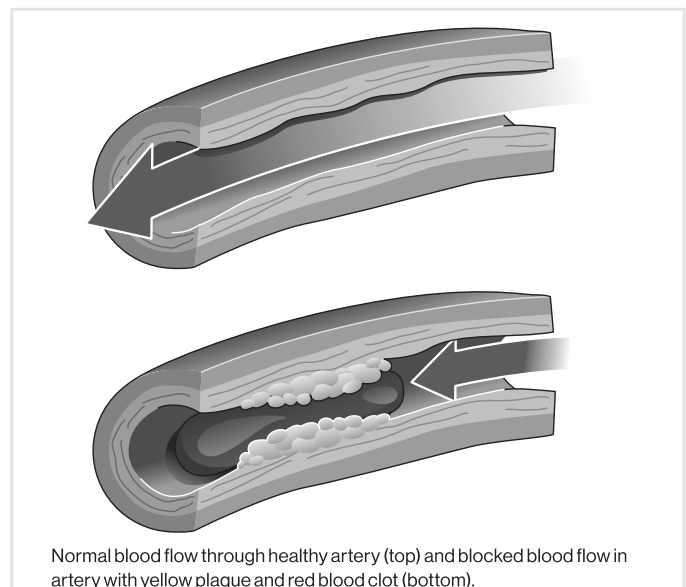
Heart attack

What is a heart attack?

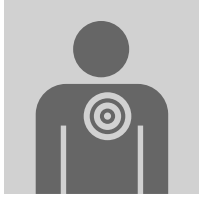
A heart attack (**myocardial infarction**) happens when the blood flow through one or more of the coronary arteries is blocked. The blockage stops oxygen from reaching parts of the heart. Not getting oxygen is what damages that part of the heart. The faster you seek medical attention and the artery is unblocked, the less damage will be done.

A heart attack can be caused in three different ways:

- 1. Atherosclerosis.** Plaque build-up can block a coronary artery so that little or no blood gets through. At the time of a heart attack, a small break in the plaque or a small blood clot (thrombus) often occurs.
- 2. Blood clot (thrombus).** Plaque is fragile and blood flow in an artery can break it open. The body tries to mend the break by making a clot. A blood clot in a coronary artery can block blood flow and cause a heart attack.
- 3. Coronary artery spasm.** Sometimes coronary arteries go into spasm (squeeze closed or contract strongly, then relax). This can stop the flow of blood through the artery and can cause a heart attack. Often, the causes of spasms are not known.

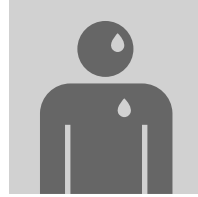


What are the common signs of heart attack?



Chest discomfort

Pressure, squeezing, fullness or pain, burning or heaviness

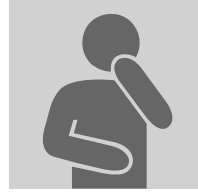


Sweating

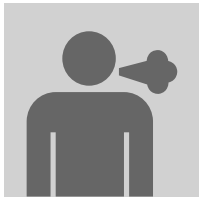


Upper body discomfort

Neck, jaw, shoulder, arms, back



Nausea



Shortness of breath



Light-headedness

© 2017- 2019 Heart and Stroke Foundation of Canada

If you are experiencing any of these signs, you should:

- **CALL 9-1-1 or your local emergency number immediately** or have someone call for you.
- Stop all activity and sit or lie down, in whatever position is most comfortable.
- If you take nitroglycerin, take your normal dosage.
- Chew and swallow one 325 mg tablet or two 81 mg tablets of ASA (Aspirin), as long as you are not allergic or intolerant.
 - Do not take other pain medications such as acetaminophen (Tylenol), or ibuprofen (Advil) instead of Aspirin.
 - Do not substitute Aspirin for medical care.
- Rest and wait for Emergency Medical Services (EMS) to arrive.
- Keep a list of your medications in your wallet and by the phone. Emergency personnel will want this information.

How is angina different from a heart attack?

Angina is not a heart attack, but angina may develop into a heart attack. The main difference between them is heart damage. Angina does not cause damage or death of the heart muscle, but a heart attack does. That is why you must act right away if you think you are having a heart attack!

	Angina	Heart attack
When does it happen?	More often during physical effort (exercise) or stress, or if you are in a very cold place, or after a large meal	Often at rest
What does it feel like?	• Pain or discomfort may spread to chest, neck, jaw, shoulders, both arms (mostly the left arm), and back • Chest tightness, burning, heaviness, feeling of squeezing or not being able to breathe • People with diabetes may only have trouble breathing	
What are other symptoms?	Sometimes dizziness, paleness, weakness	Often nausea, throwing up (vomiting), weakness, tiredness, sweating
How long does it last?	Three to five minutes, rarely more than 15 minutes	Mostly more than 30 minutes
What helps or relieves it?	Rest, nitroglycerin	Nitroglycerin or rest do not help
What happens to the heart?	No damage to the heart muscle	Small or large scar on heart muscle. This damage may be permanent if you wait too long before getting to the hospital for medical help.
What are the causes?	Short or temporary lack of oxygen to the heart	Lack of oxygen to the heart for a long time. This is usually caused by a blood clot that blocks a coronary artery.

Exercise & Blood Sugars

People taking insulin or oral diabetes pills that cause your pancreas to make more insulin are at risk for hypoglycemia with exercise; therefore:

- Carry fast acting glucose with you in case you have symptoms of low blood sugar. Common symptoms of mild to moderate low blood sugar can be found in the chart below.
- Ideally exercise within 1-2 hours of a main meal, as blood sugar peaks about 90 minutes after a meal
- Wear your Medic-Alert bracelet
- Blood sugars can drop for hours after exercise, so monitor for low blood sugar symptoms.
- Avoid exercising too close to bed time.
- **Measure blood sugars before and after exercise. Measure during exercise if you have symptoms of low blood sugar.**

Before Exercise	(less than) <5.5 mmol/L	Eat a carbohydrate-based snack and make sure glucose is above 5.5 mmol/L before starting exercise.
During Exercise	Symptoms of mild to moderate low blood sugar: • Unusual sweating • Weakness, shakiness & nervousness • Extreme hunger • Dizziness & headache • Blurred vision • Unusually fast heart beat & feeling anxious • Inability to concentrate • Confusion or irritability • Unsteadiness when standing or walking • Muscle twitching • Personality changes, such as anger or crying	Stop exercising and take 20-30 grams of fast acting glucose or 1 juice box and wait 10 minutes. If still symptomatic have another 15-20mg of glucose, every 10-15 minutes until symptoms have resolved. Probably best to head straight home or have someone pick you up. Have a snack when you get home. If you're at home, test your blood sugars and follow the guidelines for "After Exercise".
After Exercise	(less than) <4.0 mmol/L &/or Symptoms of low blood sugar	Take 20-30 grams of fast acting glucose (see examples below). Re-test after 10-15 minutes. Repeat fast acting glucose in doses of about 15-20mg, every 10-15 minutes, until glucose is over 5.0 mmol/L, then have a carbohydrate-based snack or regular meal.
	4.0-5.0 mmol/L	Even if you don't have symptoms, eat a carbohydrate-based snack if it's going to be awhile until your next meal, or have your meal. Continue to monitor for symptoms low blood sugar.

Examples of fast acting glucose:

- 1 glucose tablet = 4 grams glucose
- Hard candies (for example Life Savers) that have ~4 grams of sugar per candy
- 1 glucose gel tube = 15 grams per gel tube
- 8 ounces (1 cup) of juice or regular soda (not diet) = 20-30 grams of sugar depending on the juice
- 1 tablespoon of sugar or honey = 16 grams of sugar.

If you generally have blood sugars >5.5 mmol/L before and >5.0 mmol/L after exercise and no symptoms of low blood sugar, you may not need to test as frequently. If you're consistently having low blood sugars during or after exercise – let your Case Manager know and talk to your physician as your diabetes medications may require adjustment.

More information about symptoms of low blood sugars can be found at:

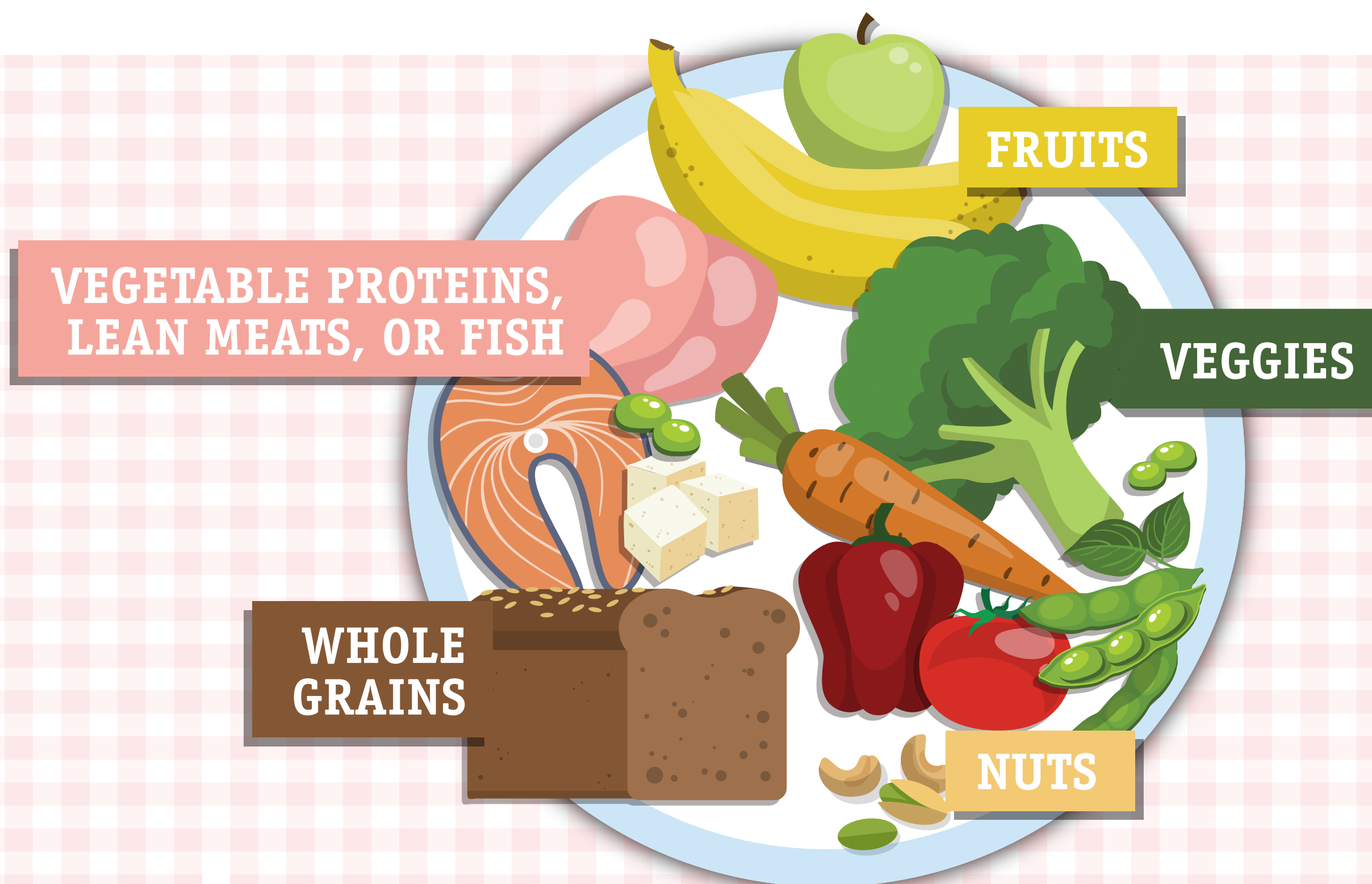
<https://www.healthlinkbc.ca/health-topics/aa20831>

More information about diabetes can be found at:

<https://www.diabetes.ca/>

<https://www.healtheuniversity.ca/en/DiabetesCollege/>

To **PROTECT YOUR HEART**, eat more...



MEDITERRANEAN



- ↓ Light on dairy, meats and sweets
- ✓ Fish
- ✓ Olive oil
- ✓ Wine (in moderation)

VEGETARIAN

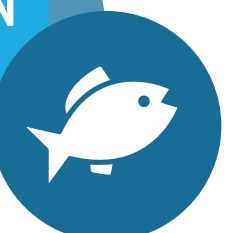


- ✗ Cuts out meat
- ✓ Vegetable proteins (soy products, legumes)

VEGAN goes further with no meat, fish, eggs or dairy products

WHOLE FOOD PLANT-BASED EATING avoids processed foods

DIETARY APPROACHES TO STOP HYPERTENSION (DASH)



- ↓ Limits salt to less than 1,500 mg/day
- ✓ Lean meat, poultry, fish
- ✓ Fruits and vegetables rich in potassium (bananas and leafy greens)

CUT BACK
on processed meats
and saturated fats

AVOID
trans fats

DRINK WATER
instead of sodas
or juice!

**MORE
TIPS**

For more information, visit [CardioSmart.org/Nutrition](https://www.cardiosmart.org/Nutrition)

 @CardioSmart

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